The value of the correct answer: a sham baseline

Simulation 2 (asynchronous activation), setup `task003-symmetric`

LLM-free simulator; report of 16 October 2026; runs of 15 October 2026

Measures of `blackboard_efficiency_info_goal_and_resource_handoff.md`, from the runs of the main report of the same date.

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the 48 most recent posts on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved) silences it, posts the one fact from its pool that best supports its target, at most 30 messages per episode. Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true. The agents hold, together, a proof of A0; no single agent does, and neither pool can prove its target (Table 1).

Table 1: Facts available at $t = 0$ and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	18 distinct	(1, 0, 0): proves A0
A0 pool (controller with target A0)	12, of which 10 new to the agents	(0.963, 0.037, 0)
A2 pool (controller with target A2)	12, of which 10 new to the agents	(0.114, 0, 0.886)
agents' facts + either pool	28 distinct	(1, 0, 0)

The runs. Posts read by an agent $q \in \{3, 6, 12\}$; posts read by the controller $q_c \in \{3, 6, 12, 24\}$; controller rate $\lambda_c \in \{0.5, 1, 2, 3, 4, 5, 6, 7, 8\}$ (one fact per message); memory $\rho \in \{1, 0.75, 0.5, 0.25\}$; vote gate on and off; proof stop on; 1,000 episodes per setting (1,740 settings). Unless stated otherwise: $q = 6$, $q_c = 12$, $\lambda_c = 1$ (the main setting, which posts about as many facts as Simulation 1 with one fact per night), gate on. The vote gate is on throughout.

Summary. We compare a controller that is given the correct answer (*informed*) with a *sham* controller that intervenes in the same way but is given a target drawn at random, using the uncontrolled swarm as the reference. **The paired correctness susceptibility χ_{correct} and the two information numbers $I(Z;X)$ and $I(Z;Y)$ behave like the susceptibility of the main report**: they grow as memory shortens (Figures 2–4), reverse with the posts each agent reads (Figure 5) and are transient with full memory (Figure 1). **The ratios η_{info} and η_{goal} do not measure size**: across settings at $t = 30$, η_{info} is uncorrelated with χ_{correct} (correlation -0.02), and with full memory it climbs to 0.93–0.95 just as every effect fades to zero. **The per-fact return η_{post} is the informative cost measure**: it falls steadily with the rate (0.05–0.06 to 0.014–0.018 per fact at $t = 30$), and it rises with the posts the agents and the controller read because the informed controller then posts fewer facts, not because the gain is larger.

1 Quantities

Treatment. An episode-level treatment Z is drawn with $P(Z = 1) = P(Z = 0) = 1/2$:

$$Z = \begin{cases} 1, & \text{informed: the controller targets the correct answer, A0;} \\ 0, & \text{sham: the controller's target is drawn by a fair coin from \{A0, A2\}, fixed for the episode.} \end{cases} \tag{1}$$

The sham intervenes like the informed controller (same rate, budget, gate and reading) but is not given the correct answer. This setup has pools for A0 and A2 only, so the sham draws from these two targets (the handoff suggests all three answers; an A1 target would need a new pool). The sham coincides with the informed controller in half of its episodes, by design. The *uncontrolled* swarm (no controller) is the reference.

State and correctness. X_t is the vote of a randomly chosen agent at time t (agents who have not voted are left out); $Y_t = \mathbf{1}\{X_t = \text{A0}\}$ says whether that vote is correct. With $x^{(e)}_{a,t}$ the share of votes for a in episode e,

$$p_{z,t}(x) = \mathbb{E}\big[x^{(e)}_{x,t} \,\big|\, Z = z\big], \qquad q_t(x) = \mathbb{E}^0\big[x^{(e)}_{x,t}\big], \qquad \bar{p}_t = \tfrac{1}{2}p_{1,t} + \tfrac{1}{2}p_{0,t}, \tag{2}$$

where $\mathbb{E}^0$ is over uncontrolled episodes. The sham's distribution is exactly the average of the A0-target and A2-target arms, $p_{0,t} = \frac{1}{2}(p^{\text{A0}}_t + p^{\text{A2}}_t)$, because each arm's distribution is an average over its episodes. Every episode was run in all three arms with the same random draws.

Paired correctness susceptibility and its return per fact.

$$\chi_{\text{correct}}(t) = \mathbb{E}\big[m^{(e)}_1(t) - m^{(e)}_0(t)\big] = \tfrac{1}{2}\,\mathbb{E}\big[x^{(e),\text{A0}}_{\text{A0},t} - x^{(e),\text{A2}}_{\text{A0},t}\big], \tag{3}$$

with $m^{(e)}_z$ the share of correct votes in episode e under treatment z. In the notation of the main report it is $\frac{1}{2}(G^{+,0}_t - G^{-,0}_t)$, the directional half-contrast $D^{(0)}_t$: it adds the help of the informed controller and the harm of the sham. With F_t the facts posted by the informed controller up to t (only its posts are counted),

$$\eta_{\text{post}}(t) = \chi_{\text{correct}}(t) \,/\, \mathbb{E}[F_t], \tag{4}$$

undefined below half a fact.

Information and the common cost. With $\text{KL}(p\|r) = \sum_x p(x)\log_2[p(x)/r(x)]$ (bits),

$$C_X(t) = \tfrac{1}{2}\text{KL}(p_{1,t}\|q_t) + \tfrac{1}{2}\text{KL}(p_{0,t}\|q_t) = I(Z;X_t) + \text{KL}(\bar{p}_t\|q_t), \tag{5}$$

$$I(Z;X_t) = \tfrac{1}{2}\text{KL}(p_{1,t}\|\bar{p}_t) + \tfrac{1}{2}\text{KL}(p_{0,t}\|\bar{p}_t), \qquad \eta_{\text{info}}(t) = I(Z;X_t)/C_X(t), \tag{6}$$

$$I(Z;Y_t) = \tfrac{1}{2}\text{kl}(r_{1,t}\|\bar{r}_t) + \tfrac{1}{2}\text{kl}(r_{0,t}\|\bar{r}_t), \qquad \eta_{\text{goal}}(t) = I(Z;Y_t)/C_X(t), \tag{7}$$

where $r_{z,t} = p_{z,t}(\text{A0})$, $\bar{r}_t = \bar{p}_t(\text{A0})$ and kl is the KL divergence between two-valued distributions. C_X is the change from no control caused by intervening at all; $I(Z;X)$ is the part of it that depends on whether the controller had the correct answer; the rest, $\text{KL}(\bar{p}\|q)$, does not. Because Y is computed from X,

$$0 \le I(Z;Y_t) \le I(Z;X_t) \le C_X(t), \qquad 0 \le \eta_{\text{goal}} \le \eta_{\text{info}} \le 1. \tag{8}$$

Both ratios are undefined when $C_X < 0.01$ bits. Every distribution gets the same pseudo-count of 0.5 episodes per value before any KL, as in the main report. In the data, Eq. (5) holds to 3×10^{-16} bits and Eq. (8) holds everywhere. Neither information nor ratio has a sign: report them with χ_{correct}.

Relation to the main report. There Z was the controller's target (A0 against A2) and C compared the two targeted arms with silence. Here Z is whether the target was the correct answer. The two information numbers are closely related (correlation 0.98 across settings at $t = 30$), but the new ones are about a third as large, because the sham shares half of its episodes with the informed controller.

2 Observables as time series

Each figure shows one memory ρ; top row: χ_{correct} (bands: 95% intervals; 0 at $t = 0$), η_{post}, η_{info}, η_{goal}; bottom row: $I(Z;X)$, $I(Z;Y)$ and C_X. One curve per controller rate λ_c; $q = 6$, $q_c = 12$, gate on; the controller stops at $t = 30$ (dashed). Axes are the same in all four figures.

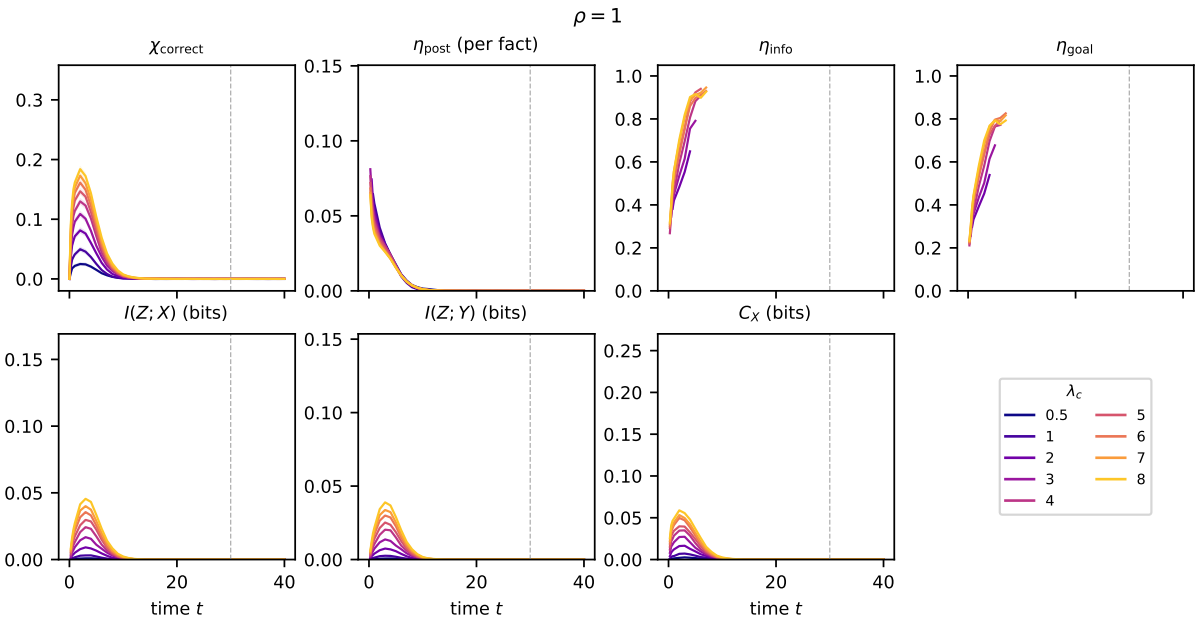


Figure 1: Full memory, $\rho = 1$. Blank: undefined ($C_X < 0.01$ bits, or under half a fact posted).

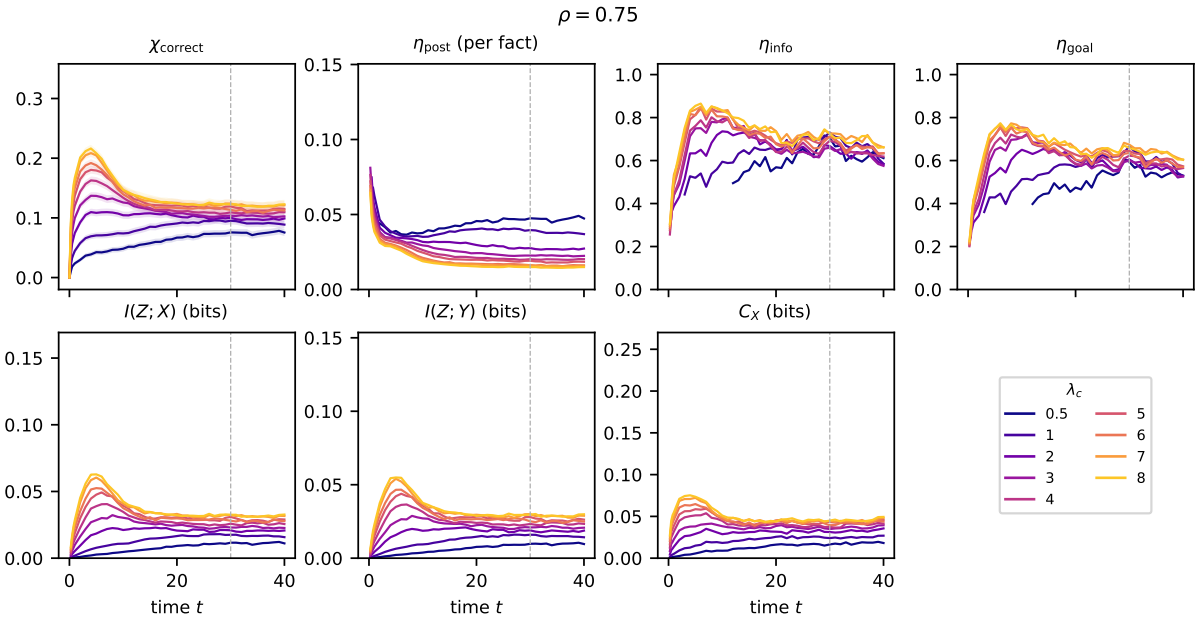


Figure 2: $\rho = 0.75$.

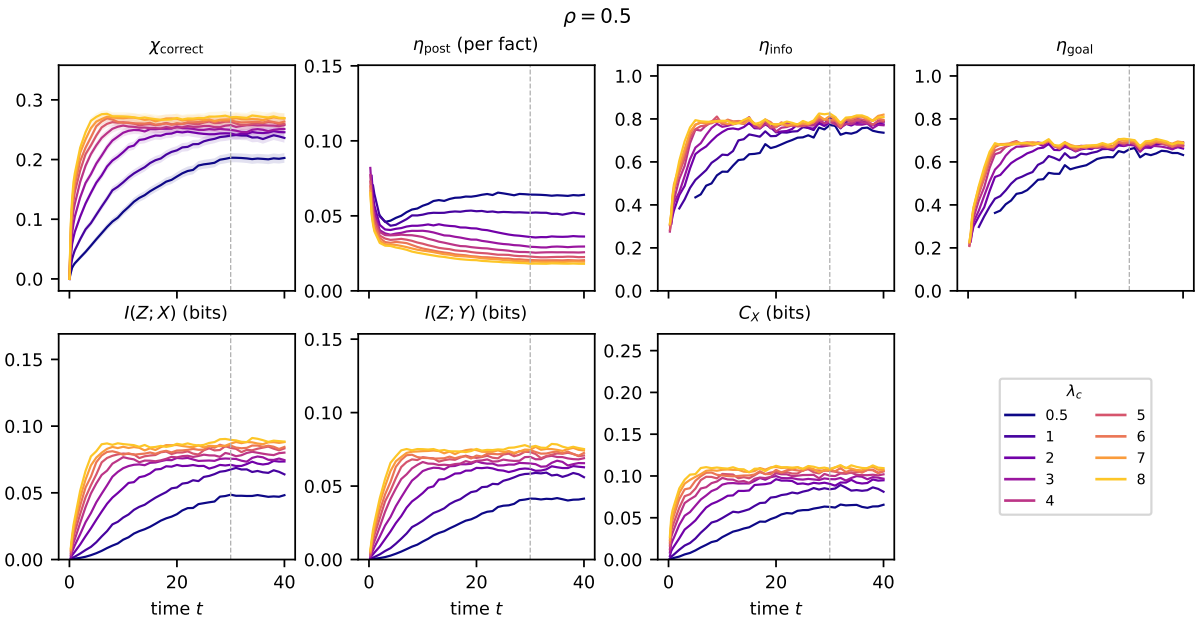


Figure 3: $\rho = 0.5$.

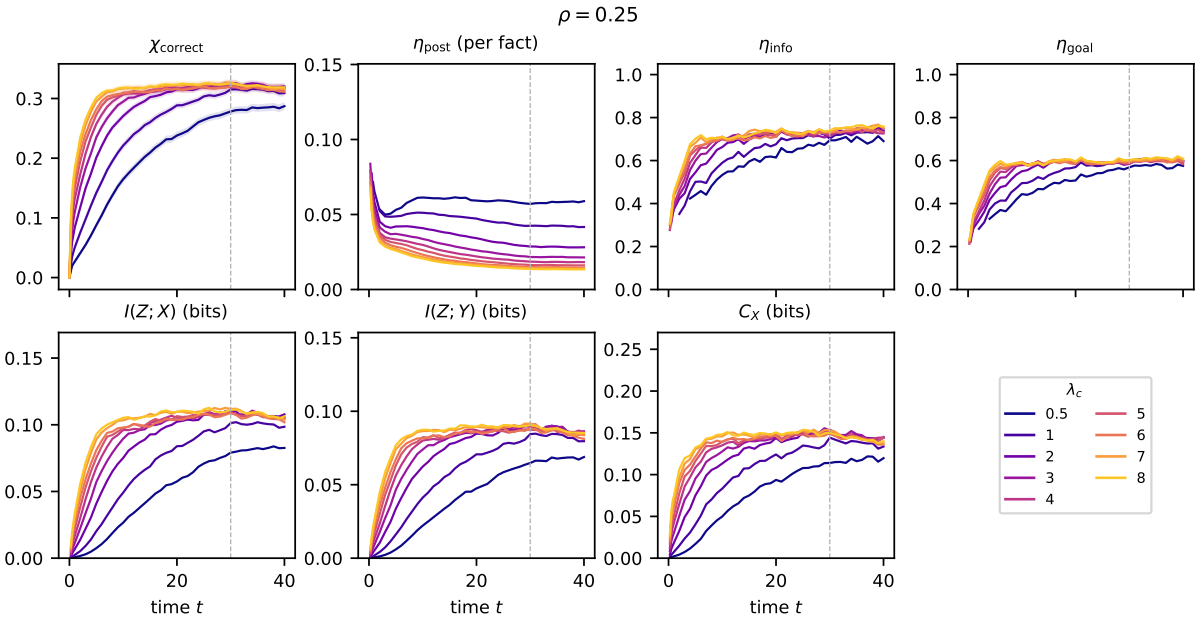


Figure 4: $\rho = 0.25$.

- χ_{correct} **grows as memory shortens**: at $t = 30$, 0.08–0.12 ($\rho = 0.75$), 0.20–0.27 ($\rho = 0.5$), 0.28–0.33 ($\rho = 0.25$), 0 with full memory, where it peaks at 0.18 ($t \approx 2$, $\lambda_c = 8$) and vanishes by $t \approx 10$.
- $I(Z;X)$, $I(Z;Y)$ **and** C_X **have the same shape as** χ_{correct}: they rise, overshoot where it overshoots, and persist after $t = 30$ ($I(Z;X) = 0.01$–0.03, 0.05–0.09, 0.08–0.11 bits at $t = 30$ for $\rho = 0.75, 0.5, 0.25$).
- **Most of the information is about correctness**: $I(Z;Y)/I(Z;X) = 0.82$–0.90 at $t = 30$ for $\rho < 1$. The rest is in how the wrong votes split between A1 and A2.
- η_{info} **and** η_{goal} **rise early and then flatten**: from 0.37–0.53 and 0.29–0.44 at $t = 1$ to 0.66–0.83 and 0.57–0.70 at $t = 30$, at every $\rho < 1$ and for every rate.
- **With full memory the ratios climb as the effects vanish**: η_{info} reaches 0.93–0.95 just before C_X falls below 0.01 bits (Figure 1). A ratio of two vanishing numbers can be large; it says nothing about how much control there is.

- η_{post} **falls with the rate and with time**: every controller starts near 0.04–0.07 per fact; at $t = 30$ the slowest keeps 0.05–0.06, the fastest 0.014–0.018 (all $\rho < 1$).

3 Phase diagrams

All panels at $t = 30$, gate on; colour scales are shared by the three figures. Grey number in χ_{correct}: its 95% interval includes 0. Dash: undefined.

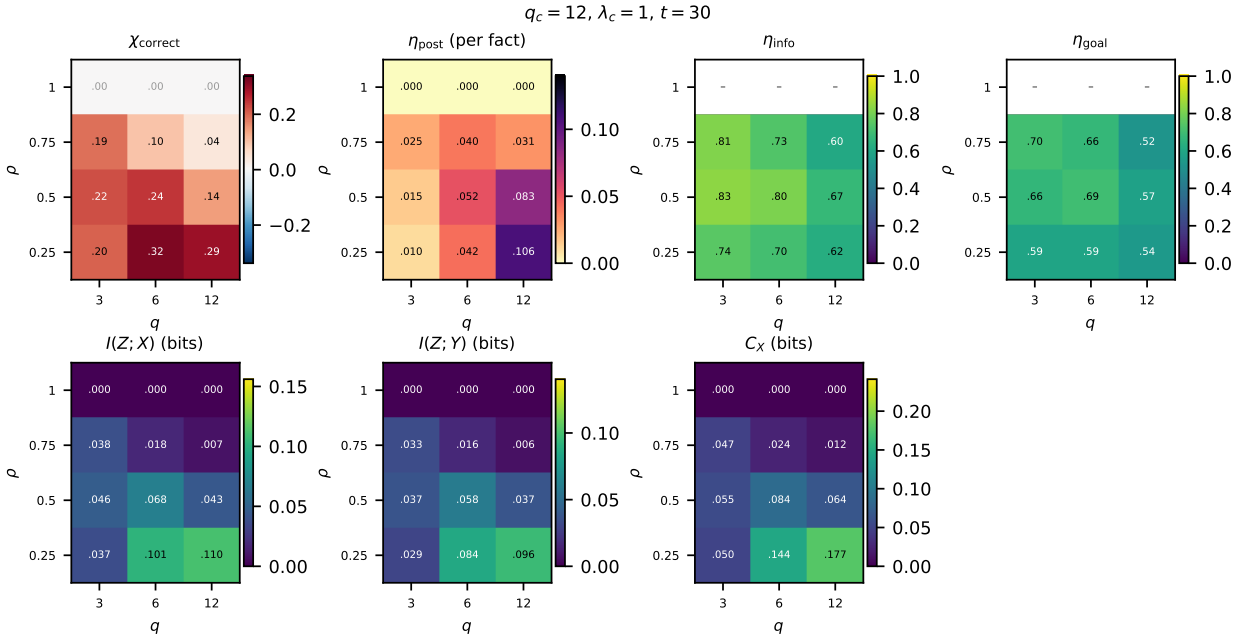


Figure 5: Over memory ρ and the posts each agent reads q; $q_c = 12$, $\lambda_c = 1$.

- **The reversal with q is in χ_{correct} and the information numbers**: at $\rho = 0.75$, χ_{correct} falls from 0.19 ($q = 3$) to 0.04 ($q = 12$) and $I(Z;X)$ from 0.038 to 0.007 bits; at $\rho = 0.25$ they rise from 0.20 to 0.29–0.32 and from 0.037 to 0.10–0.11.
- η_{info} **and** η_{goal} **do not show it**: they fall with q at every ρ (η_{info} 0.81 → 0.60 at $\rho = 0.75$; 0.74 → 0.62 at $\rho = 0.25$).
- η_{post} **rises with** q (0.010 → 0.106 per fact at $\rho = 0.25$) because the informed controller posts fewer facts when agents read more (19 against 3 by $t = 30$), likely because its gate finds agreement sooner (not checked).

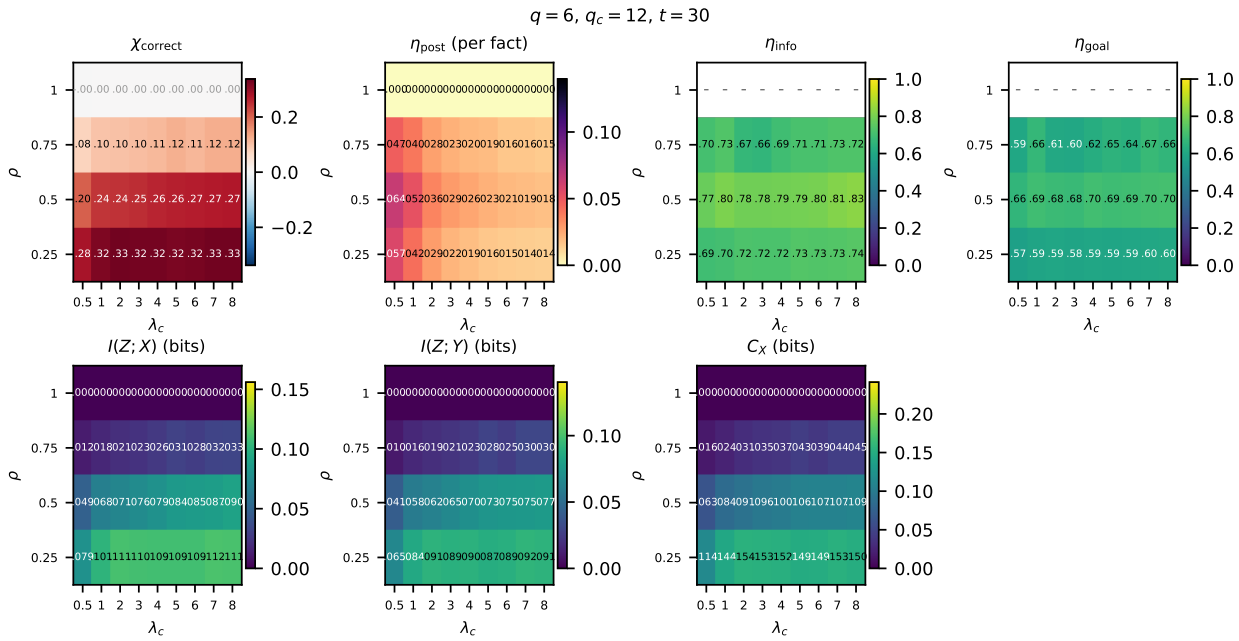


Figure 6: Over memory ρ and the controller rate λ_c; $q = 6$, $q_c = 12$.

- **Memory sets χ_{correct}, $I(Z;X)$, $I(Z;Y)$ and C_X; the rate barely does**: along each row they change little beyond $\lambda_c = 1$, as the main report's susceptibility does.
- η_{info} **and** η_{goal} **stay at** 0.66–0.83 **and** 0.57–0.71 over the whole diagram, highest at $\rho = 0.5$, with no relation to the size of χ_{correct}.
- η_{post} **falls along every row**, from 0.05–0.06 at $\lambda_c = 0.5$ to 0.014–0.018 at $\lambda_c = 8$: faster control adds facts but little correctness.

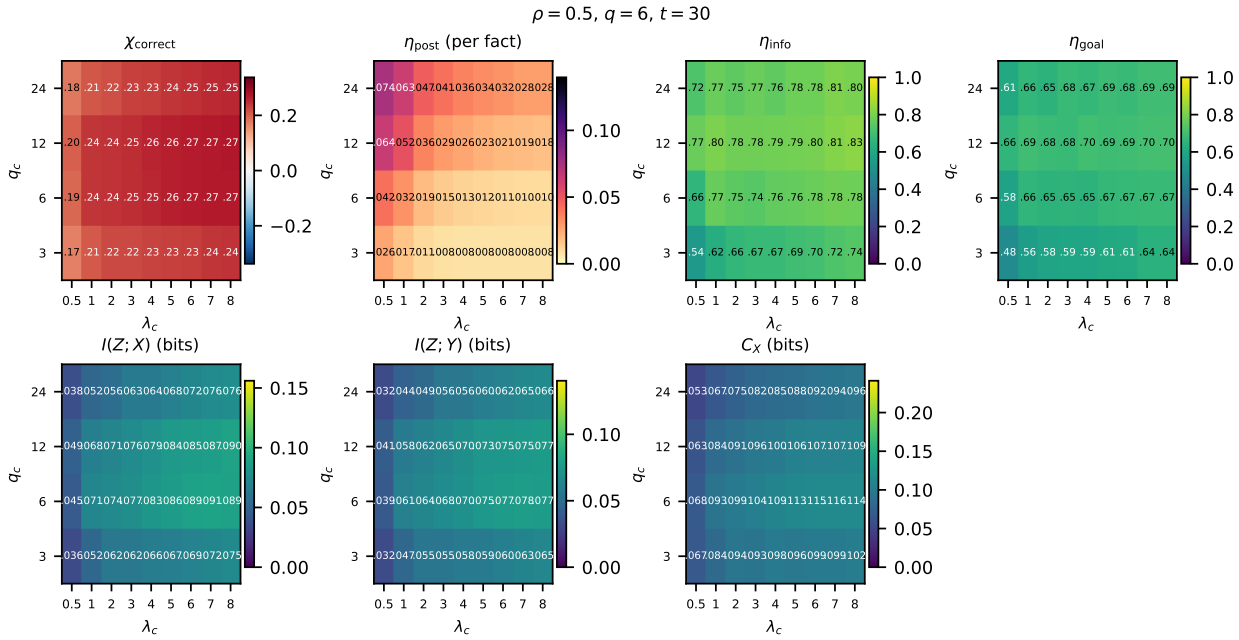


Figure 7: Over the posts the controller reads q_c and its rate λ_c; $\rho = 0.5$, $q = 6$.

- **The controller's reading hardly changes** χ_{correct} (mean 0.22, 0.25, 0.25, 0.23 for $q_c = 3, 6, 12, 24$) **but quadruples** η_{post} (0.011 → 0.043 per fact), because it posts fewer facts (24 at $q_c = 3$, 6 at $q_c = 24$, by $t = 30$), likely because reading more lets its gate stop it sooner (not checked).
- η_{info} **and** η_{goal} **rise slightly with** q_c (0.67 → 0.77–0.80 and 0.59 → 0.67–0.69).

4 Conclusions

1. **The sham baseline gives a clean signed measure**: χ_{correct} follows the size of control as the main report's susceptibility does (correlation with $I(Z;X)$ across settings: 0.90). It mixes the informed controller's help with the sham's harm, so it should be read next to the gain against no control.
2. η_{info} **and** η_{goal} **are not efficiencies of control.** Across settings they are uncorrelated with χ_{correct} (-0.02), they hide the reversal with q, and they approach 1 as every effect vanishes. Their numerators and C_X carry the information; the ratios describe only how the change is split, as the handoff warns.
3. η_{post} **is the useful cost measure**, with one caution: it rises whenever the informed controller posts less, so a setting that silences the controller sooner (more reading by agents or controller) looks more efficient per fact even when χ_{correct} does not change.
4. **Most target information is about correctness** ($I(Z;Y)/I(Z;X) = 0.82$–0.90).

Caveats. Simulation 2, `task003-symmetric`, one world; a sham over the two targets that have pools; 1,000 episodes per arm. Reads are not charged: the handoff's η_{ops} would add them with a weight fixed in advance. Data: `results/simulation_2/2026-10-15_sim2_symmetric_fine`; table: `results/gold_standard/2026-10-15/sham_tabl` code: `analysis/task003_llm_free/shared_analysis/gold_standard/sham_baseline*.py`.